

Improving a Machine Learning Calibration Framework for Low-Cost PM_{2.5} Sensors Through Methodological Enhancements

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Abstract—Machine learning models have become one of the most widely used approaches for calibrating low-cost PM_{2.5} sensors. However, their performance strongly depends on the methodological decisions adopted during model development. This paper analyzes a recently proposed adaptive calibration framework and evaluates different strategies to improve its performance, including an ablation study, the incorporation of meteorological variables, the use of alternative machine learning algorithms, and hyperparameter optimization through Optuna. The evaluation was conducted using a network of six low-cost sensors deployed at three locations in the Metropolitan Region of Greater Vitória, Brazil, with two sensors co-located with an air quality reference station for calibration purposes. The results identified the adaptive trust mechanism as the component with the greatest influence on the framework performance, while the comparison of different machine learning algorithms showed XGBoost to be the best-performing alternative. The proposed methodology achieved MAE reductions greater than 91% for all evaluated sensors, reaching a maximum improvement of 92.6%, while also reducing RMSE and eliminating the systematic bias. These findings indicate that specific methodological modifications can significantly improve the performance of existing calibration frameworks for low-cost air quality sensors.

Index Terms—Low-cost PM_{2.5} sensors, Machine learning, Sensor calibration, XGBoost.

I. INTRODUCTION

AIR pollution remains one of the major environmental and public health challenges worldwide. Among the different atmospheric pollutants, fine particulate matter (PM_{2.5}) has become one of the most extensively studied air pollutants due to its ability to penetrate deep into the respiratory system and even enter the bloodstream, increasing the risk of respiratory, cardiovascular, and other chronic diseases [1], [2]. Consequently, continuous monitoring of PM_{2.5} concentrations has become an essential tool for assessing air quality, protecting public health, and supporting environmental policy and decision-making.

Traditionally, air quality monitoring relies on reference-grade stations equipped with highly accurate instruments capable of providing reliable measurements under standardized operating conditions. However, the high costs associated with the acquisition, installation, operation, and maintenance of these systems considerably limit the number of available

monitoring stations. As a result, the spatial coverage of monitoring networks is reduced, making it difficult to implement monitoring systems in regions with limited resources or in areas where reference stations are unavailable, thereby limiting access to reliable air quality information [3]. In fact, the World Health Organization reports that monitoring networks remain concentrated mainly in major urban areas, leaving significant gaps in air quality monitoring coverage, particularly in rural and underserved regions [4].

In response to these limitations, low-cost sensors have attracted increasing attention as a complementary alternative to conventional monitoring networks. Their main advantage lies in enabling the deployment of a larger number of monitoring points at a substantially lower cost, improving the spatial and temporal resolution of air quality data [5] while supporting applications such as smart cities, distributed environmental monitoring, and citizen science. Despite these advantages, measurement accuracy remains one of the main challenges associated with their use, as sensor responses can be influenced by meteorological conditions [5], sensor drift, component aging, and manufacturing variability [6]. For this reason, calibration has become a fundamental step in ensuring the reliability of the collected data.

To improve measurement quality, numerous calibration methodologies have been proposed in the literature. Early studies mainly relied on statistical models and linear regression techniques [5], [7], [8]. With the advancement of machine learning, models capable of representing nonlinear relationships between sensor responses, meteorological variables, and reference measurements have been extensively developed. Algorithms such as Random Forest, Support Vector Regression, and Gradient Boosting, among others, have enabled improvements in calibration performance across a wide range of applications [9], [10], [11], [12], [13]. Similarly, Artificial Neural Networks have also been successfully applied for sensor calibration tasks [14]. However, these models often rely on highly complex architectures whose decision-making processes are difficult to interpret and are therefore frequently regarded as *black-box* approaches, particularly in the case of deep neural networks.

Within this context, [15] proposed a dynamic calibration framework based on trust-weighted consensus mechanisms. Unlike conventional approaches, where each sensor is calibrated independently, the proposed methodology combines information from the reference station with the individual measurements of each sensor and the information shared by

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neighboring sensors, weighting the contribution of each source through a dynamic trust index. In this way, the calibration process takes advantage of both local and distributed information across the network, resulting in a model that is relatively simple, interpretable, and easy to implement without relying on highly complex machine learning algorithms.

Motivated by the advantages of this strategy, this work proposes a series of methodological modifications aimed at improving its performance while preserving the simplicity and interpretability that characterize the original framework. To this end, a pilot study was conducted using a network of six low-cost $PM_{2.5}$ sensors and a reference station employed for model training and evaluation. The original methodology was first implemented on this experimental infrastructure and subsequently subjected to a systematic methodological assessment to identify the contribution of its main components and explore potential opportunities for improvement. Based on these findings, a modified calibration framework was developed and evaluated under identical experimental conditions. Finally, both approaches were compared using multiple performance metrics to assess the impact of the proposed modifications on the calibration performance.

II. METHODOLOGY

A. Experimental Setup

The experimental campaign was conducted in the Metropolitan Region of Greater Vitória (RMGV), Espírito Santo, Brazil, using a pilot monitoring network composed of six low-cost $PM_{2.5}$ sensors distributed across three monitoring locations, with two sensors co-located at the reference station and the remaining four deployed in pairs at two additional sites. Figure 1 illustrates the geographical distribution of the monitoring locations together with the reference station.

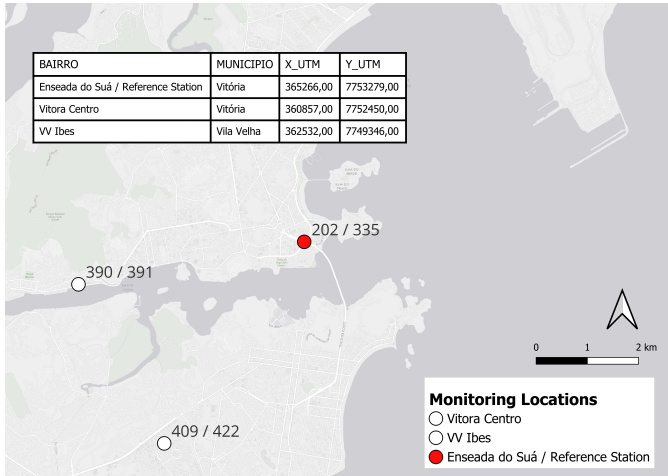


Fig. 1. Geographical distribution of the monitoring locations and the reference station in RMGV.

Each sensing unit was part of the QAr embedded monitoring platform developed by the Centro de Pesquisa, Inovação e Desenvolvimento (CPID). Although the platform is capable of measuring multiple environmental variables, this study focused exclusively on $PM_{2.5}$ concentrations obtained using the

Sensirion SPS30 optical particulate matter sensor. In addition, temperature, relative humidity, and geographic coordinates were also recorded. Temperature and relative humidity were incorporated into the methodological assessment, while geographic coordinates were used to identify neighboring sensors within the monitoring network.

Reference measurements were obtained from the RGV4 Enseada do Suá air quality monitoring station operated by the Instituto Estadual de Meio Ambiente (IEMA). The experimental campaign was carried out from June 26 to July 3, 2026, with measurements recorded every minute, resulting in approximately 9,300 observations per sensor. The resulting dataset served as the basis for implementing the original calibration framework, performing the methodological assessment, and evaluating the proposed improvements.

TABLE I
SUMMARY OF THE EXPERIMENTAL SETUP

Parameter	Description
Study area	Metropolitan Region of Greater Vitória, Brazil
Monitoring locations	3
Number of sensors	6
$PM_{2.5}$ sensor	Sensirion SPS30
Reference station	IEMA RGV4 – Enseada do Suá
Sampling interval	1 min
Variables	$PM_{2.5}$, temperature, relative humidity, GPS coordinates
Observations	~9,300 per sensor

B. Original Calibration Framework

The calibration framework proposed by [15] was adopted as the baseline methodology in this study. To ensure a fair comparison, the original implementation provided by the authors was reproduced without modifications and directly applied to the experimental dataset described in the previous section. The framework combines information from the reference station, individual sensor measurements, and neighboring sensors through a trust-based consensus strategy. Its objective is to improve the accuracy of low-cost $PM_{2.5}$ sensors while maintaining a relatively simple and interpretable calibration process. Since a detailed description of the algorithm and its mathematical formulation is available in [15], only the main stages are summarized here.

The calibration process begins with an offset-scale correction intended to remove the main systematic errors affecting each sensor. A linear regression is fitted between the raw sensor measurements and the corresponding reference observations to estimate an offset (a_s) and a scaling factor (b_s). The corrected measurement is then obtained as

$$x'_s(t) = b_s (x_s(t) - a_s), \quad (1)$$

where $x_s(t)$ represents the original sensor measurement and $x'_s(t)$ the corrected value. This initial correction provides a common baseline before the remaining calibration stages are applied.

Once the systematic errors have been reduced, the methodology assigns a trust score to each sensor based on four complementary performance indicators: accuracy, stability, responsiveness, and consensus with neighboring sensors. These indicators are combined into a single trust value according to

$$T_s = \frac{A_s + S_s + R_s + C_s}{4}, \quad (2)$$

where A_s , S_s , R_s , and C_s denote the normalized scores associated with each criterion. This trust score determines the confidence assigned to each sensor and is subsequently used to guide the calibration process.

The corrected measurements are then transformed into an extended feature set combining temporal, statistical, and spatial information. The original framework incorporates wavelet-based descriptors, rolling statistical features, time-of-day information, and a spatial consensus term computed from neighboring sensors. These features are used to train an individual Gradient Boosting Machine (GBM) model for each sensing unit, whose hyperparameters are adjusted according to the corresponding trust score. Finally, the calibrated concentration is obtained through a weighted combination of the corrected measurement and the prediction generated by the machine learning model,

$$\hat{y}_s(t) = \omega(T_s)x'_s(t) + [1 - \omega(T_s)]M_s(F_s(t)), \quad (3)$$

where $M_s(\cdot)$ represents the sensor-specific GBM model, $F_s(t)$ denotes the extracted feature vector, and $\omega(T_s)$ is a weighting function that depends on the trust score.

Although the original framework demonstrates good calibration performance, the contribution of its individual components is not evaluated independently in the original study. Consequently, it remains unclear which stages have the greatest impact on the overall calibration performance and whether some of them could be simplified or replaced while preserving, or even improving, the model accuracy. Motivated by this observation, the following section presents a systematic methodological assessment of the original framework and the set of modifications proposed in this work.

C. Proposed Methodological Improvements

The proposed methodological improvements were developed through a three-stage evaluation process. First, an ablation study was performed to evaluate the individual contribution of the main components of the original model. Next, several alternative calibration strategies were investigated, including the incorporation of additional input variables, the evaluation of different machine learning algorithms, and the analysis of alternative calibration configurations. Finally, based on the evidence obtained from these experiments, an improved calibration framework was defined and subsequently refined through hyperparameter optimization.

1) *Ablation Study*: The original calibration framework was first analyzed through an ablation study. Three independent configurations were evaluated by selectively disabling specific components while preserving the remaining stages of the

methodology unchanged. This procedure made it possible to isolate the effect of each component on the overall calibration performance.

In the first configuration, the adaptive trust weighting mechanism was disabled by assigning a constant trust value of 0.5 to all sensors. Consequently, every sensor was treated with the same confidence level throughout the calibration process. This experiment was designed to assess whether the dynamic trust weighting proposed in the original framework provided a measurable benefit over a fixed weighting strategy.

The second configuration removed all wavelet-derived features while preserving the remaining statistical, temporal, and spatial variables used by the model. This evaluation aimed to determine the contribution of multiscale information to the calibration performance.

Finally, the third configuration removed the spatial consensus feature, preventing the calibration model from incorporating information from neighboring sensors. This experiment assessed the importance of exploiting the spatial relationships among sensors within the monitoring network.

The performance of each configuration was subsequently compared with that of the complete implementation of the original framework using the evaluation metrics described in Section III.

2) Evaluation of Alternative Calibration Strategies:

a) *Meteorological Variables*: To evaluate the influence of environmental conditions on the calibration performance, air temperature and relative humidity were incorporated as additional predictor variables during the feature engineering stage. This modification was implemented while preserving the overall structure of the original framework, allowing the contribution of the meteorological variables to be evaluated independently.

The inclusion of these variables was motivated by previous studies reporting that the response of low-cost optical particulate matter sensors can be significantly influenced by meteorological conditions, particularly relative humidity, due to hygroscopic particle growth and changes in light-scattering behavior [5], [6]. Consequently, it was hypothesized that explicitly incorporating these environmental variables would allow the model to better represent these effects and, potentially, improve the calibration performance.

b) *Reference Station Dependency*: To evaluate the dependence of the calibration framework on the reference station, an alternative configuration was implemented in which the reference measurements were completely removed from the calibration process. In this configuration, both the initial offset-scale correction and the trust score computation were performed using only the information provided by the low-cost sensor network.

Likewise, the spatial consensus obtained from neighboring sensors was used as the target variable during model training. This configuration made it possible to evaluate whether the information shared among the sensors was sufficient to maintain an adequate calibration performance in the absence of a reference station.

c) *Machine Learning Models*: To evaluate the influence of the learning algorithm on the calibration performance, four regression models were compared: Gradient Boosting Machine (GBM), adopted in the original framework, Random Forest (RF), Extreme Gradient Boosting (XGBoost), and Linear Regression (LR). The original feature set and the overall structure of the calibration framework were preserved, so that any differences in performance could be attributed to the learning algorithm itself.

All configurations were applied to the same dataset and compared using the mean absolute error (MAE), computed both for each individual sensor and for the average response of the sensor network. This evaluation guided the selection of the learning algorithm that was subsequently incorporated into the final proposed framework.

3) *Proposed Calibration Framework*: Based on the methodological analyzes described in the previous sections, a modified calibration framework was defined. The proposed methodology preserves the overall structure of the original framework while incorporating a set of modifications derived from the evaluation of its individual components and the alternative calibration strategies investigated.

In this version, the original adaptive trust mechanism was removed and replaced by a fixed configuration throughout the calibration process. In addition, the input feature set was expanded by incorporating air temperature, relative humidity, and two temporal lag features corresponding to one- and two-minute previous corrected sensor readings. These variables were included as additional features during the feature engineering stage.

The Gradient Boosting Machine adopted in the original methodology was replaced by XGBoost. Furthermore, the calibration model was reformulated using a residual learning strategy, in which the algorithm predicts the difference between the corrected sensor measurement and the corresponding reference measurement. The final calibrated concentration was then obtained by adding a fraction of the predicted residual to the corrected sensor reading, according to

$$\hat{y}_s(t) = x'_s(t) + 0.9 \hat{e}_s(t), \quad (4)$$

where $x'_s(t)$ denotes the sensor measurement after the offset-scale correction stage, and $\hat{e}_s(t)$ represents the residual error estimated by the XGBoost model.

Finally, once the proposed configuration had been established, an automatic hyperparameter optimization was performed using Optuna. The optimization aimed to minimize the average MAE across the sensor network by tuning the main XGBoost hyperparameters over 30 optimization trials. The resulting parameter values were subsequently incorporated into the final version of the the final calibration framework.

III. RESULTS AND DISCUSSION

A. Original Calibration Framework

The original calibration methodology was implemented using the dataset described in Section II-A to evaluate its

performance on an independent dataset from that employed in the original study [15]. Table II summarizes the main performance metrics obtained for each sensor before and after the calibration process.

TABLE II
PERFORMANCE OF THE ORIGINAL CALIBRATION FRAMEWORK

Sensor	MAE		RMSE		Bias	
	Orig.	Cal.	Orig.	Cal.	Orig.	Cal.
202	3.415	1.923	5.023	2.513	-2.073	-0.002
335	3.352	1.963	5.098	2.561	-1.728	-0.003
390	3.446	2.020	5.676	2.633	-1.601	-0.002
391	3.718	1.928	6.277	2.549	-0.906	-0.004
409	3.350	1.915	4.909	2.506	-1.916	-0.002
422	3.479	1.951	5.470	2.541	-1.756	-0.003

In general, the original methodology produced a consistent reduction in prediction errors across all six evaluated sensors. The MAE decreased from values ranging between 3.35 and $3.72 \mu\text{g}/\text{m}^3$ to values between 1.91 and $2.02 \mu\text{g}/\text{m}^3$, while the RMSE also showed substantial reductions in every case. Likewise, the systematic bias observed in the original measurements was almost completely removed after calibration, resulting in bias values close to zero for all sensors.

These results indicate that the framework proposed by [15] maintains a robust performance when applied to an independent dataset, successfully reducing both systematic and random errors in low-cost $\text{PM}_{2.5}$ sensor measurements. Nevertheless, the remaining prediction errors and the differences observed among sensors suggest that there is still room for further methodological improvements.

To complement the quantitative evaluation presented in Table II, Fig. 2 illustrates the temporal response of the six sensors before and after applying the original calibration framework, together with the corresponding reference measurements.

As shown in Fig. 2, the calibration process enables the sensor measurements to more closely follow the temporal evolution recorded by the reference station, particularly during periods of higher variability in $\text{PM}_{2.5}$ concentrations. Although small discrepancies remain during some time intervals, the original framework consistently improves the agreement between the low-cost sensors and the reference measurements across all evaluated sensors.

B. Ablation Study

In order to identify the individual contribution of the main components of the original framework, the three ablation scenarios described in Section II-C were evaluated by independently removing the trust mechanism, the wavelet features, and the spatial consensus component, while keeping the remainder of the methodology unchanged. Table III summarizes the MAE obtained for each sensor under the different evaluated scenarios.

The results show that all three ablation scenarios reduced the MAE with respect to the complete original framework, indicating that some components introduce additional complexity that, for the dataset analyzed in this study, does not necessarily translate into improved calibration performance.

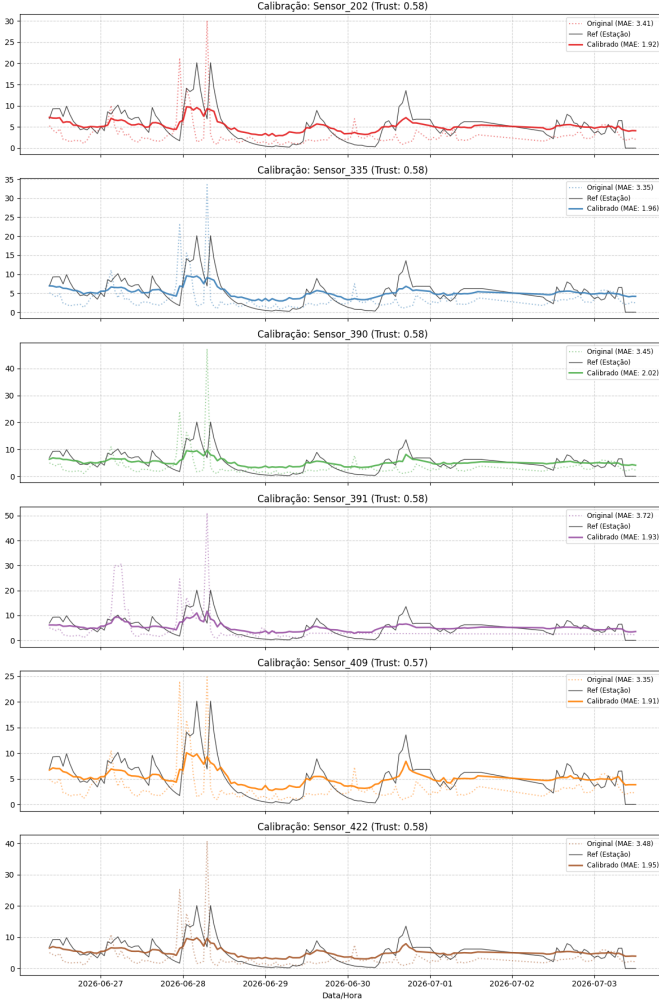


Fig. 2. Temporal response of the six low-cost $PM_{2.5}$ sensors before and after applying the original calibration framework.

TABLE III
PERFORMANCE COMPARISON OF THE ABLATION STUDY SCENARIOS

Sensor	Uncal.	Original Framework	Without Trust	Without Wavelets	Without Consensus
202	3.415	1.923	0.861	1.071	1.097
335	3.352	1.963	0.872	1.098	1.133
390	3.446	2.020	0.861	1.198	1.172
391	3.718	1.928	0.848	1.110	1.148
409	3.350	1.915	0.820	1.084	1.089
422	3.479	1.951	0.830	1.066	1.107

Among the evaluated components, removing the trust mechanism produced the largest reduction in MAE for all six sensors, yielding the lowest errors among all the analyzed scenarios. This result suggests that the adaptive weighting strategy adopted by the original framework did not provide additional benefits to the calibration process for the dataset considered in this study. Furthermore, it also indicates that the adaptive trust mechanism may not generalize equally well across different datasets, motivating the evaluation of alternative trust configurations in the proposed methodology.

The removal of the wavelet features and the spatial con-

sensus component also resulted in lower errors than those obtained with the complete framework, although in both cases the performance remained slightly inferior to that achieved by removing the trust mechanism. These findings indicate that both components continue to provide useful information for the calibration process, although their contribution is smaller than the impact introduced by the adaptive trust mechanism.

Overall, the ablation study identified the trust mechanism as the main opportunity for improvement within the original framework. This finding served as the basis for the methodological modifications investigated in the following sections.

C. Evaluation of Alternative Calibration Strategies

a) *Meteorological Variables:* To evaluate the influence of meteorological conditions on the calibration process, air temperature and relative humidity were incorporated as additional input features during the feature engineering stage while keeping the remainder of the original framework unchanged. Tables IV and V summarize the resulting calibration performance and trust scores, respectively.

TABLE IV
CALIBRATION PERFORMANCE AFTER INCORPORATING METEOROLOGICAL VARIABLES

Sensor	MAE Original	MAE Calibrated	Reduction (%)
202	3.415	1.967	42.41
335	3.352	1.978	40.98
390	3.446	2.027	41.18
391	3.718	1.898	48.94
409	3.350	1.935	42.23
422	3.479	1.979	43.10

Contrary to expectations, incorporating meteorological variables did not improve the calibration performance with respect to the original framework. Instead, slightly higher MAE values were obtained for most of the evaluated sensors, despite the additional information provided to the learning model.

TABLE V
TRUST SCORES OBTAINED AFTER INCORPORATING METEOROLOGICAL VARIABLES

Sensor	Trust Score
202	0.606
335	0.607
390	0.611
391	0.591
409	0.594
422	0.608

A comparison between both configurations also reveals a consistent increase in the trust scores, with values ranging from 0.591 to 0.611 for all sensors. A possible explanation is that these higher trust values led the adaptive framework to select less complex model configurations during training, limiting its ability to correct the remaining sensor errors. This observation suggests that the effectiveness of meteorological variables depends not only on the additional information they provide, but also on how they interact with the adaptive mechanisms of the calibration framework.

b) *Reference Station Dependency*: To evaluate the possibility of reducing the dependence on a reference station, the measurements from the reference station were replaced by the spatial consensus of neighboring sensors during the offset-scale correction, trust score computation, and model training stages. Table VI summarizes the calibration performance obtained under this configuration.

TABLE VI
CALIBRATION PERFORMANCE WITHOUT A REFERENCE STATION

Sensor	MAE	MAE	Reduction (%)
	Original	Calibrated	
202	3.415	3.352	1.86
335	3.352	3.343	0.27
390	3.446	3.275	4.98
391	3.718	3.157	15.10
409	3.350	3.345	0.14
422	3.479	3.320	4.55

Unlike the supervised approach, the reference-free configuration produced only marginal improvements over the uncalibrated measurements, indicating that spatial consensus alone was insufficient to accurately calibrate the sensor network. These results show that, for the evaluated framework, information provided by a reference station remains essential to achieve accurate calibration.

c) *Machine Learning Models*: To evaluate the influence of the machine learning algorithm on the calibration framework, the original Gradient Boosting model was compared with Random Forest (RF), Linear Regression (LR), and XGBoost while keeping the remainder of the methodology unchanged. Although the original study adopted a Gradient Boosting model, the rationale for this choice was not explicitly discussed. Therefore, alternative machine learning algorithms widely used for regression were evaluated.

Figure 3 compares the MAE obtained for each sensor using the different machine learning models.

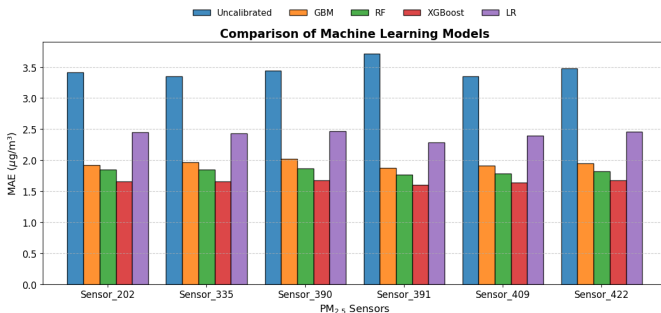


Fig. 3. Comparison of the MAE obtained by different machine learning models for each evaluated sensor.

The results show that XGBoost consistently achieved the lowest MAE for all six evaluated sensors, with values ranging from 1.60 to 1.67 $\mu\text{g}/\text{m}^3$, outperforming the original Gradient Boosting model as well as Random Forest and Linear Regression. Although Random Forest also improved the calibration performance compared with Gradient Boosting, XGBoost consistently provided the largest error reduction across all sensors.

Based on these results, XGBoost was selected as the regression model for the proposed methodology.

D. Proposed Calibration Framework

Based on the findings obtained from the previous analyses, the proposed calibration framework was evaluated using the optimal hyperparameters obtained through Optuna. Table VII presents the final configuration selected for the XGBoost model.

TABLE VII
OPTIMIZED XGBOOST HYPERPARAMETERS

Hyperparameter	Value
Number of Trees	202
Learning Rate	0.089
Maximum Depth	6
Subsample	0.713
Column Sample by Tree	0.900

Table VIII summarizes the performance achieved by the proposed methodology. Compared with the uncalibrated measurements, the final framework reduced the MAE by more than 91% for all evaluated sensors, reaching a maximum improvement of 92.6%. Likewise, the RMSE was substantially reduced, while the systematic bias was completely eliminated, demonstrating the ability of the proposed framework to accurately correct both random and systematic measurement errors.

TABLE VIII
PERFORMANCE EVALUATION OF THE PROPOSED CALIBRATION FRAMEWORK

Sensor	MAE		RMSE		Bias	
	Orig.	Cal.	Orig.	Cal.	Orig.	Cal.
202	3.415	0.283	5.023	0.371	-2.073	0.000
335	3.352	0.284	5.098	0.372	-1.728	0.000
390	3.446	0.285	5.676	0.373	-1.601	0.000
391	3.718	0.274	6.277	0.362	-0.906	0.000
409	3.350	0.283	4.909	0.371	-1.916	0.000
422	3.479	0.284	5.470	0.372	-1.756	0.000

Finally, Fig. 4 compares the original and calibrated time series with the reference station measurements. The calibrated signals closely follow the reference throughout the entire monitoring period, providing a visual confirmation of the substantial improvements achieved by the proposed methodology.

IV. CONCLUSIONS

The performance of an adaptive machine learning framework for the calibration of low-cost $\text{PM}_{2.5}$ sensors was analyzed to identify opportunities for improving its methodology. Based on this analysis, a set of methodological modifications was proposed, combining a fixed trust mechanism, temporal lag features, meteorological variables, XGBoost, and Optuna-based hyperparameter optimization.

The experimental results showed that the evaluated modifications do not contribute equally to the calibration process. The ablation study identified the adaptive trust mechanism as the component with the greatest influence on the overall performance, whereas the inclusion of meteorological variables

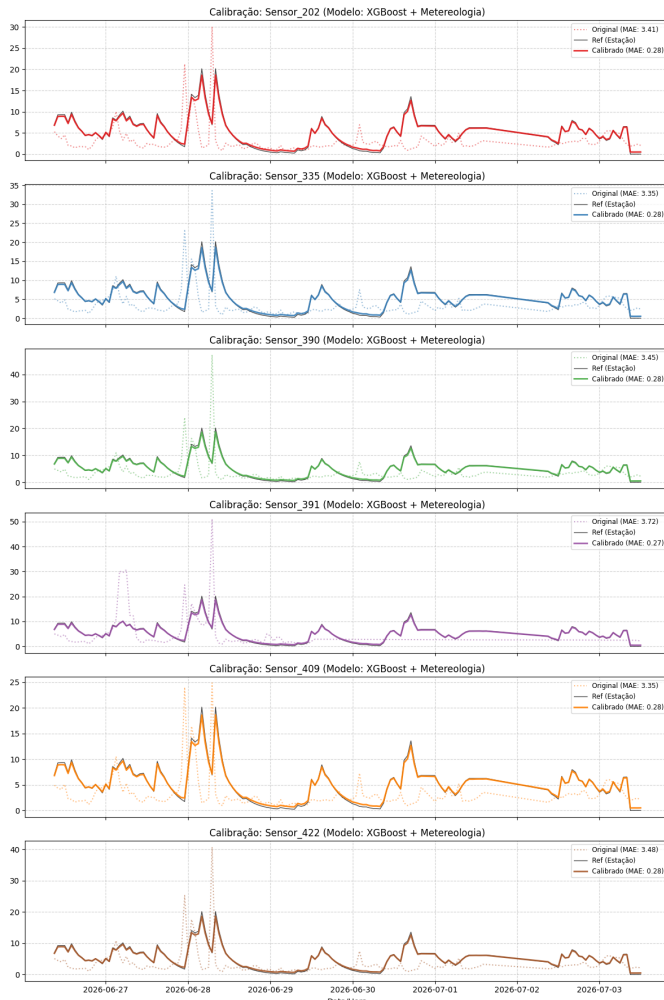


Fig. 4. Temporal response of the six low-cost $PM_{2.5}$ sensors before and after applying the proposed calibration framework.

did not improve the calibration for the evaluated dataset. In addition, the experiments confirmed that measurements from a reference station remain essential for achieving accurate sensor calibration.

The proposed methodology reduced the MAE by more than 91% for all six evaluated sensors, reaching a maximum improvement of 92.6%. RMSE was also substantially reduced, while the systematic bias was eliminated after calibration. These findings indicate that combining relatively simple methodological modifications can substantially improve the performance of existing calibration frameworks for low-cost air quality sensors.

Future work will focus on validating the proposed methodology using longer monitoring campaigns, different environmental conditions, and other low-cost sensor technologies to further assess its robustness and generalization capability.

REFERENCES

- [1] G. D. Thurston, H. Kipen, I. Annesi-Maesano, J. Balmes, R. D. Brook, K. Cromar, S. De Matteis, F. Forastiere, B. Forsberg, M. W. Frampton *et al.*, "A joint ers/ats policy statement: what constitutes an adverse health effect of air pollution? an analytical framework," *European Respiratory Journal*, vol. 49, no. 1, 2017.
- [2] D. E. Schraufnagel, "The health effects of ultrafine particles," *Experimental & molecular medicine*, vol. 52, no. 3, pp. 311–317, 2020.
- [3] B. Mijling, H. Eskes, S. Hofmann, P. Moreno, D. García Falin, and M. E. de Vega Pastor, "High-resolution mapping of urban no2 concentrations using retina v2: a case study on data assimilation of surface and satellite observations in madrid," *Geoscientific Model Development*, vol. 18, no. 18, pp. 6439–6460, 2025.
- [4] World Health Organization, *WHO global air quality guidelines: particulate matter (PM_{2.5} and PM₁₀), ozone, nitrogen dioxide, sulfur dioxide and carbon monoxide*. Geneva: World Health Organization, 2021.
- [5] A. Ilenič, A. Mauko Pranjić, J. Ščančar, R. M. Ščančar, and P. Kumar, "Can low-cost sensors (lcs) enhance air quality monitoring for personal pollution exposure assessment?" *Environmental Research Communications*, vol. 7, no. 9, p. 095007, 2025.
- [6] M. R. Giordano, C. Malings, S. N. Pandis, A. A. Presto, V. McNeill, D. M. Westervelt, M. Beekmann, and R. Subramanian, "From low-cost sensors to high-quality data: A summary of challenges and best practices for effectively calibrating low-cost particulate matter mass sensors," *Journal of Aerosol Science*, vol. 158, p. 105833, 2021.
- [7] X. Li, C. M. Mak, Y. Dai, K. W. Ma, and H. M. Wong, "Hyper-localized pollution mapping using low-cost wearable monitors and citizen science in hong kong," *Buildings*, vol. 15, no. 17, p. 3131, 2025.
- [8] J. Hofman, B. Lazarov, C. Stroobants, E. Elst, I. Smets, and M. Van Poppel, "Portable sensors for dynamic exposure assessments in urban environments: state of the science," *Sensors*, vol. 24, no. 17, p. 5653, 2024.
- [9] R. Xu, D. Yao, Y. Pian, R. Cao, Y. Fu, X. Yang, T. Gan, and Y. Liu, "Integrating mobile and fixed monitoring data for high-resolution pm_{2.5} mapping using machine learning," *arXiv preprint arXiv:2503.12367*, 2025.
- [10] V. Sooriyaarachchi, D. J. Lary, L. O. Wijeratne, and J. Waczak, "Causality-driven feature selection for calibrating low-cost airborne particulate sensors using machine learning," *Sensors*, vol. 24, no. 22, p. 7304, 2024.
- [11] V. Kumar, M. Sahu, B. Biswal, J. Prakash, S. Choudhary, R. Raliya, T. S. Chadha, J. Fang, and P. Biswas, "Real-time source apportionment of particulate matter from low-cost particle sensors using machine learning," *Aerosol science and engineering*, vol. 9, no. 4, pp. 446–456, 2025.
- [12] A. Wang, Y. Machida, P. deSouza, S. Mora, T. Duhl, N. Hudda, J. L. Durant, F. Duarte, and C. Ratti, "Leveraging machine learning algorithms to advance low-cost air sensor calibration in stationary and mobile settings," *Atmospheric Environment*, vol. 301, p. 119692, 2023.
- [13] R. T. Wahyuni, D. Hanafi, M. R. Tomari, and D. S. S. Sahid, "Spatial estimation of pm_{2.5} using random forest with temporally adjusted portable sensors, land use, and aod in a data-scarce urban area," *International Journal of Intelligent Engineering & Systems*, vol. 18, no. 10, 2025.
- [14] L. L. Sachetti, E. B. Cussuol, J. M. S. Nogueira, and V. F. Mota, "pmsensing: A participatory sensing network for predictive monitoring of particulate matter," *arXiv preprint arXiv:2111.11441*, 2021.
- [15] S. Mahajan and D. Helbing, "Dynamic calibration of low-cost pm_{2.5} sensors using trust-based consensus mechanisms," *npj Climate and Atmospheric Science*, vol. 8, no. 1, p. 257, 2025.